

Can We Have Both? Balancing Agency and Attention in Route-Guided, Focused VR Experiences

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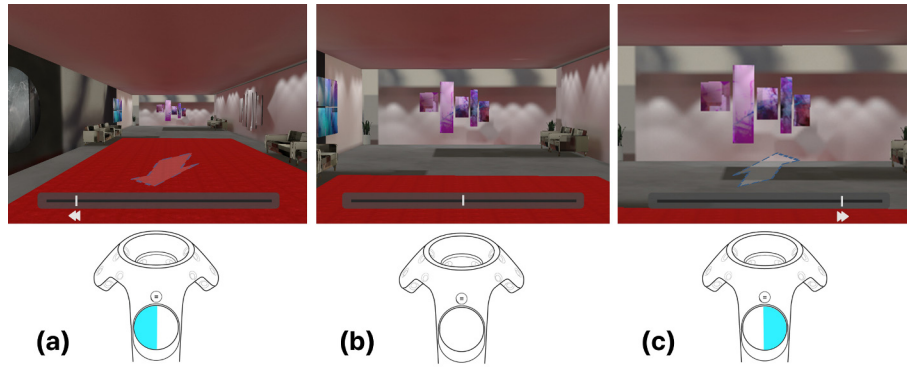


Figure 1: Semi-active locomotion as a design probe in a route-guided VR experience. It offers minimal, intentional control over progression along a predefined route: (a) pressing left moves the user backward along the traversed path, (b) releasing pauses movement, and (c) pressing right moves the user forward toward the destination. This constrained interaction reduces navigational decision-making while preserving intentional progression control, enabling the isolation of visual attention and perceived agency in route-guided VR experiences. Illustrative arrows and progress bars are not shown during actual use.

ABSTRACT

VR locomotion often entails a trade-off: active control enhances user agency but can divert attention from content, whereas passive travel supports focused viewing at the cost of autonomy. We investigated whether semi-active control can mediate this dichotomy by comparing active, passive, and semi-active locomotion in a user study ($N = 24$) conducted in a structured, route-guided VR museum. Eye-tracking analyses showed that semi-active locomotion sustained exhibit-focused visual attention at a level comparable to passive travel, while subjective agency remained comparable to active control. These findings suggest that minimal, intentional input can effectively balance agency and attention in focused, predefined VR experiences.

Index Terms: Virtual Reality; Locomotion; Visual Attention; Sense of Agency; Eye Tracking

1 INTRODUCTION

Locomotion is a fundamental interaction in virtual reality (VR) that plays a central role in shaping user experience, including presence, agency, and attentional allocation. VR locomotion techniques are commonly located along an active-passive spectrum [1]. Active methods, such as joystick-based navigation, provide users with a strong sense of agency and presence, but they often require continuous cognitive and physical effort, which can divert attention away from primary virtual content [4]. In contrast, passive methods, such as on-rails navigation, reduce interaction effort and support observation, but frequently diminish the user's sense of agency, resulting

in a more detached or spectator-like experience [2].

However, achieving both is crucial for VR applications such as virtual exhibitions, educational environments, and training simulations, where users must maintain focus to comprehend content while retaining a sense of volition to ensure motivation and presence [3]. To address this challenge, we investigate whether this trade-off can be mediated through semi-active locomotion, a hybrid approach in which low-level steering is automated while movement initiation remains under user control. By reducing navigation input to discrete, intentional commands, this approach aims to decouple the psychological sense of intention from the cognitive burden of continuous steering. We report on a comparative user study that examined active, passive, and semi-active modalities in a structured, route-guided VR museum environment. In doing so, we explore the potential of semi-active interaction to harmonize the relationship between agency and engagement, offering an alternative design perspective beyond the traditional framing of VR locomotion.

2 EXPERIMENT

To empirically examine how different degrees of locomotion control mediate the agency-attention trade-off, we conducted a 1×3 within-subjects user study in a virtual exhibition environment that requires both user agency for exploration and sustained attention to engage with exhibits. Participants used an HTC Vive Pro Eye HMD with integrated eye tracking; locomotion input was provided via the Vive controller touchpad. Condition order was counterbalanced to minimize learning effects. Twenty-four participants (17 females and 7 males; aged = 19-29, $M = 23.04$, $SD = 2.68$) completed the study. The study was approved by the university IRB.

Locomotion Conditions. We compared three locomotion designs that represent different degrees of control along the active-passive spectrum. To control for potential confounders, the movement speed was fixed at 1.5 m/s under all conditions, and participants retained freedom of head movement for visual exploration.

Active (Joystick): Participants exercised continuous control over movement direction via touchpad input while moving at a fixed

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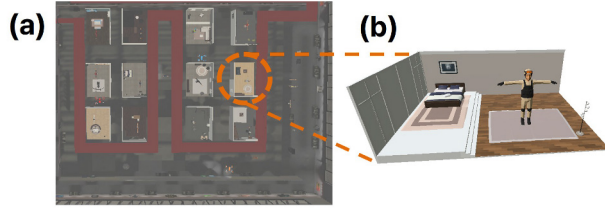


Figure 2: The virtual exhibition environment used in the study. (a) Top-down view of the exhibition layout with 12 booths and a red-carpet path guiding to minimize wayfinding load. (b) Example of an individual booth, featuring a simple interior and a human figure.

speed, representing fully user-driven navigation.

Passive (On-Rails): Participants were automatically transported along the same predefined route, with no control over movement direction or progression, representing fully system-driven navigation.

Semi-Active: Participants controlled their progression along the same predefined route using minimal, discrete input while steering was fully automated. Pressing the right region of the touchpad moved the user forward toward the destination along the upcoming path, pressing the left region moved the user backward along the already traversed path, and releasing the input paused movement (see Fig 1 for an illustration).

Environment. The study utilized a virtual exhibition consisting of independent booths arranged along a fixed route (see Fig. 2). Participants were instructed to freely observe the exhibits while navigating the hall. To approximate a realistic viewing environment and assess the robustness of visual attention, the scene included controlled visual distractors (e.g., moving avatars or drones) placed alongside, but not within, the primary exhibit areas. Also, red-carpet path guided to minimize wayfinding-related cognitive load.

Measures. The dependent measures included: (1) Subjective Agency, assessed using the Sense of Agency Scale (SoAS) [5], which measures sense of positive agency (SoPA) and sense of negative agency (SoNA) through 11 items rated on a 7-point Likert scale. (2) Visual Attention, measured using the Booth Fixation Ratio, defined as the proportion of gaze fixation time on booth-related content relative to the total viewing time in the exhibition area ($N = 21$, excluding missing eye-tracking data).

3 RESULTS

All data satisfied the assumptions of normality and homogeneity of variance (Shapiro-Wilk: all $p > .069$; Levene's test: all $p > .487$). Consequently, we conducted mixed-design ANOVA with condition as a within-subject factor and order as a between-subject factor, followed by Bonferroni-corrected paired t -tests for post-hoc comparisons. The significance level was set at $\alpha = .05$ (see Fig 3).

Subjective Agency. Significant main effects of locomotion condition were observed for both positive agency (SoPA; $F(2,36) = 36.71$, $p < .001^{***}$, $ges = 0.358$) and negative agency (SoNA; $F(2,36) = 29.10$, $p < .001^{***}$, $ges = 0.393$), while the effect of order was not significant (SoPA: $F(5,18) = 0.63$, $p = .677$; SoNA: $F(5,18) = 0.89$, $p = .511$). Post-hoc tests showed that SoPA scores were higher in the Semi-Active and Active conditions than in the Passive condition (both $p < .001^{***}$), while SoNA scores were higher in the Passive condition than in the Semi-Active and Active conditions (both $p < .001^{***}$). Semi-Active and Active yielded comparable agency ratings (SoPA: $p = .165$; SoNA: $p = 1$). Together, these results suggest that semi-active condition yields intentional control at a level comparable to active navigation, while mitigating the external control associated with passive locomotion.

Visual Attention. For visual attention (Booth Fixation Ratio), ANOVA revealed a significant main effect of locomotion condition

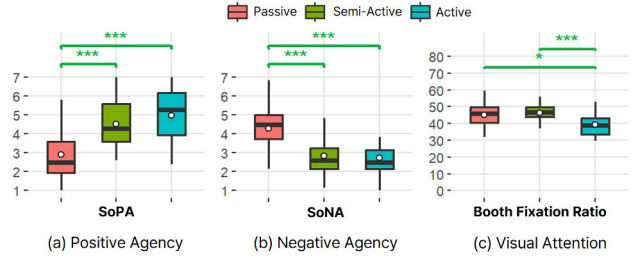


Figure 3: Results for (a) Positive Agency, (b) Negative Agency and (c) Visual Attention across locomotion conditions. White circles denote means; * $p < .05$, *** $p < .001$.

($F(2,30) = 16.61$, $p < .001^{***}$, $ges = 0.296$), whereas the effect of order was not significant ($F(5,15) = 0.43$, $p = .818$). Post-hoc analyses showed that the Booth Fixation Ratio was significantly higher in the Semi-Active condition than in the Active condition ($p < .001^{***}$). The Passive condition also exhibited a significantly higher fixation ratio than the Active condition ($p = .013^*$). No significant difference was observed between the Semi-Active and Passive ($p = 1$). These findings indicate that semi-active locomotion supports focused visual attention at a level comparable to passive.

4 CONCLUSION

This work investigated how different degrees of locomotion control shape the relationship between agency and attention in route-guided VR experiences. Through a comparative user study, we showed that semi-active locomotion occupies a meaningful middle ground between fully active and fully passive approaches. By requiring minimal, intentional input while automating low-level steering, semi-active locomotion preserved users' sense of agency at a level comparable to active navigation, while supporting exhibit-focused visual attention comparable to passive travel.

These findings indicate that the trade-off between agency and attention in VR locomotion is not strictly binary. Instead, semi-active interaction offers a practical design strategy for VR applications that require sustained focus on content, including virtual exhibitions, educational environments, and training simulations.

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